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FOURTH REVIEW UNDER THE THREE-YEAR ARRANGEMENT UNDER THE EXTENDED CREDIT FACILITY AND REQUEST FOR MODIFICATION OF PERFORMANCE CRITERIA—DEBT SUSTAINABILITY ANALYSIS UPDATE

Approved By
Markus Rodlauer and Peter Allum (IMF)
Ernesto May and Jeffrey Lewis (IDA)

Prepared by
International Monetary Fund
International Development Association

Risk of external debt distress	Low
Augmented by significant risks stemming from domestic public and/or private external debt?	No

This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA Update from November 11, 2013.[1] *The results indicate that Bangladesh remains at a low risk of external public debt distress, and with no significant vulnerabilities related to domestic debt or private external debt.*[2]

A. Underlying Assumptions

There are two major changes in the assumptions relative to the last update to the DSA:

- First, nominal GDP in U.S. dollars grows slightly faster from FY17 onward.[3] This reflects the impact of higher US inflation (from the revised *World Economic Outlook* projections) on domestic inflation and the GDP deflator. In consequence, nominal GDP in FY34 is now estimated to be about 8 percent higher.

[1] Based on the end-June 2011 stock of debt, the last full DSA was prepared in late 2011 (SM/11/278, Supplement 1 and IDA/SEC/65557). In line with the *Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries* (SM/13/292, IDA/SEC/82566), a full DSA is expected to be prepared once every three years for PRGT-eligible, IDA-only countries. In between short updates are expected to be produced unless macroeconomic conditions since the last full DSA have significantly changed.

[2] For the purposes of this DSA, the public sector comprises the central government and nonfinancial public enterprises. This analysis is based on the joint Fund-Bank debt sustainability framework for conducting debt sustainability analysis in low-income countries. Under IDA's Country Policy and Institutional Assessment (CPIA), Bangladesh is assessed to be a medium performer, with an average rating of 3.35 during 2010–12. This DSA update uses the indicative thresholds for countries for this category.

[3] Data are on a fiscal year basis unless otherwise noted; FY17 is the fiscal year July 2016–June 2017.

- Second, the balance in the General Provident Fund (GPF)—equivalent to about 2 percent of GDP at end-FY13—is now included in the DSA as government debt. The GPF holds no assets, and participants' net contributions are a source of financing for the central government (reflected in the fiscal table under nonbank financing). They therefore represent a government liability.

B. External DSA

Under the baseline scenario, Bangladesh's public and publicly guaranteed (PPG) external debt is projected to trend down slightly. All associated PPG indicators remain well within the respective policy-dependent solvency thresholds under the baseline scenario and all associated standard bound tests.

C. Public DSA

The present value (PV) of public debt is projected to remain well below the respective benchmark. This result continues to hold under all standard bounds tests.

D. Alternative Scenario

The alternative scenario—similar to the one discussed in the November 2013 DSA update—assumes the following combination of shocks to the baseline: (1) issuance of a US$1 billion (0.6 percent of GDP) 10-year sovereign bond in FY15, to be rolled over in FY25; (2) new nonconcessional borrowing of US$6 billion (2.3 percent of GDP in FY19) to construct two 1,000 megawatt nuclear reactors during FY17–21, repayment of which is assumed to be financed by additional sovereign bond issuances starting in FY24; and (3) an additional recapitalization of the state-owned banks of 1.3 percent of FY15 GDP, beyond what is already considered in the baseline. This additional recapitalization would restore the banks' tangible capital to the regulatory minimum after deducting the estimated impact on capital positions arising from the temporary relaxation in loan rescheduling rules in December 2013 and also assuming a full write-off of all nonperforming loans (NPLs) and fraudulent loans not yet classified as NPLs. This is clearly a worst-case scenario.

Under this alternative scenario, no policy-dependent sustainability thresholds are breached even when key variables are subjected to standard bound tests. Thus, the risk of external debt distress remains low also in the alternative scenario. Similarly, the PV of public debt remains below the respective solvency and liquidity thresholds.

E. Conclusion

The macroeconomic framework is broadly unchanged from the November 2013 DSA update. With no major changes to prospective borrowing assumptions (volumes and costs), the debt sustainability assessment remains unchanged and the risk of external and public debt distress continues to be classified as low.

Table 1. Bangladesh: DSA Update: Key Variables 1/

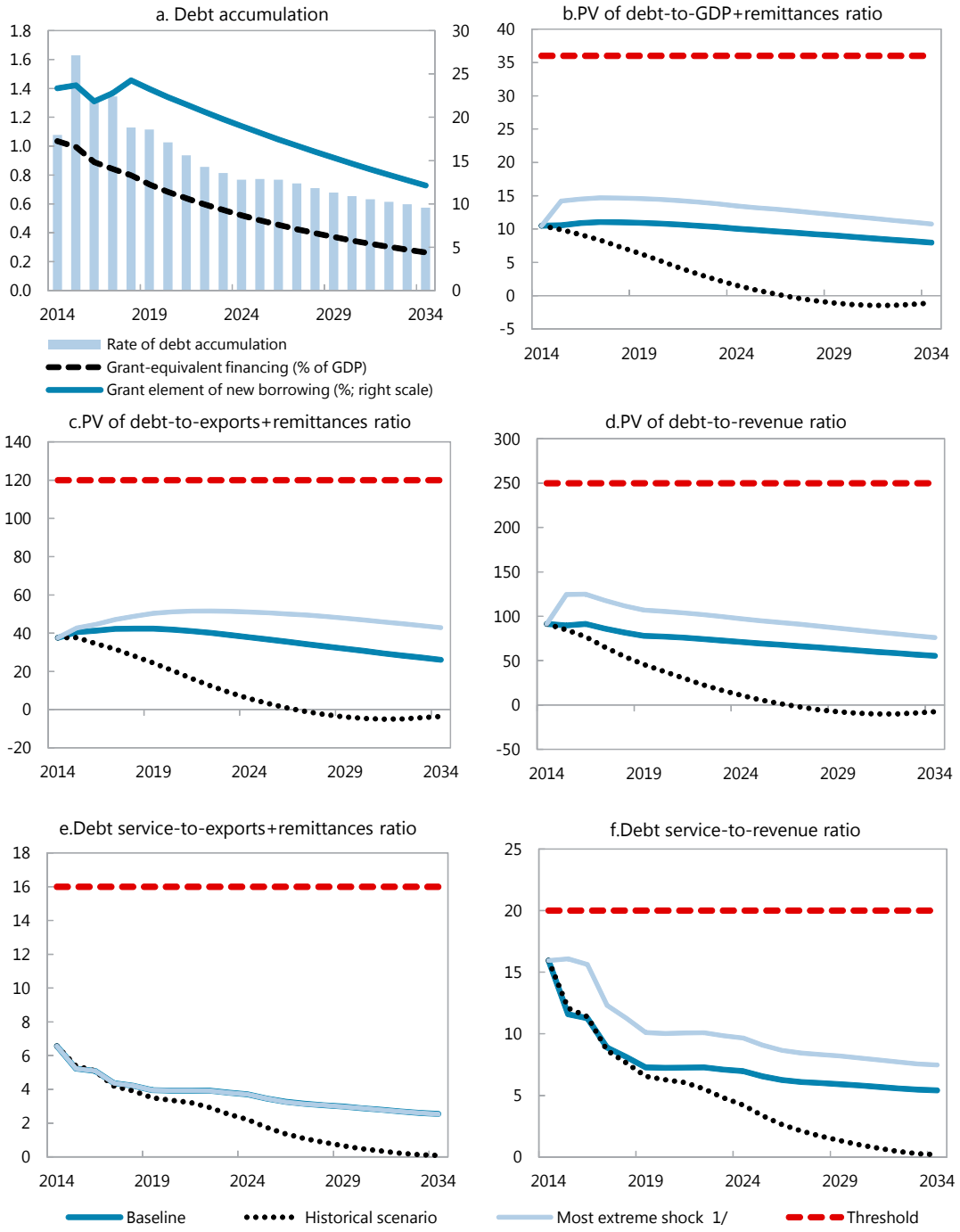
	2011	2012	2013	2014	2015	2016	2017	2018	2019	2024	2029	2034
	(Percent of GDP, unless otherwise indicated)											
Nominal GDP (US$ billions)	112	116	130	153	175	192	212	234	258	412	656	1044
Real GDP (percentage change)	6.7	6.2	6.0	5.5	6.3	6.5	7.0	7.0	7.0	6.5	6.5	6.5
GDP deflator (percentage change)	7.5	8.5	6.6	8.7	7.5	6.4	6.2	5.9	6.1	5.2	4.5	4.5
Fiscal (central government)												
Total revenue and grants	11.9	12.9	12.9	13.0	13.2	13.4	14.2	14.9	15.3	15.2	15.2	15.1
Foreign grants	0.3	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.3	0.2	0.2	0.1
Total expenditure	16.0	16.3	16.9	16.8	16.9	17.1	17.5	17.7	17.8	18.1	18.1	18.3
Interest payments	2.0	2.2	2.3	2.2	2.1	2.0	2.0	2.0	1.9	1.8	1.9	2.1
Overall balance	-4.1	-3.4	-4.0	-3.7	-3.7	-3.7	-3.3	-2.8	-2.5	-2.9	-3.0	-3.2
Primary balance	-2.2	-1.2	-1.7	-1.5	-1.6	-1.7	-1.3	-0.8	-0.6	-1.1	-1.0	-1.1
Net domestic financing	3.8	2.6	2.5	2.9	2.5	2.4	2.1	1.9	1.7	2.4	2.6	3.0
Net external financing	0.4	0.8	1.2	0.8	1.2	1.3	1.2	0.9	0.8	0.6	0.4	0.2
Balance of payments												
Exports of goods and services	22.5	23.0	22.6	21.4	19.8	20.5	20.4	20.5	20.5	22.2	24.7	27.7
Imports of goods and services	34.3	33.6	30.5	27.5	27.3	27.7	27.4	27.1	26.8	27.6	29.5	31.5
Workers' remittances	10.4	11.0	11.0	9.0	8.3	8.1	7.7	7.4	7.1	6.0	5.0	4.2
Current account, including official transfers	-2.0	-0.4	1.9	1.3	-0.6	-0.7	-0.8	-0.9	-0.9	-1.3	-1.7	-1.6
Foreign direct investment	0.7	1.0	1.3	1.0	1.0	1.0	1.1	1.1	1.1	1.6	2.1	2.5
External borrowing												
Central government	0.9	1.5	1.9	1.5	1.8	1.8	1.6	1.4	1.4	1.1	1.0	0.8
Public enterprises with guarantee	0.0	0.0	0.0	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Gross official reserves	3.0	3.1	4.3	4.8	5.0	5.1	5.2	5.2	5.2	4.9	4.7	4.9
(Months of imports of goods and services)												

Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ Data on a fiscal year basis, with 2014 corresponding to July 2013–June 2014.

Figure 1. Bangladesh: Indicators of Public and Publicly Guaranteed External Debt under Different Assumptions, 2014-2034 1/

(In percent)

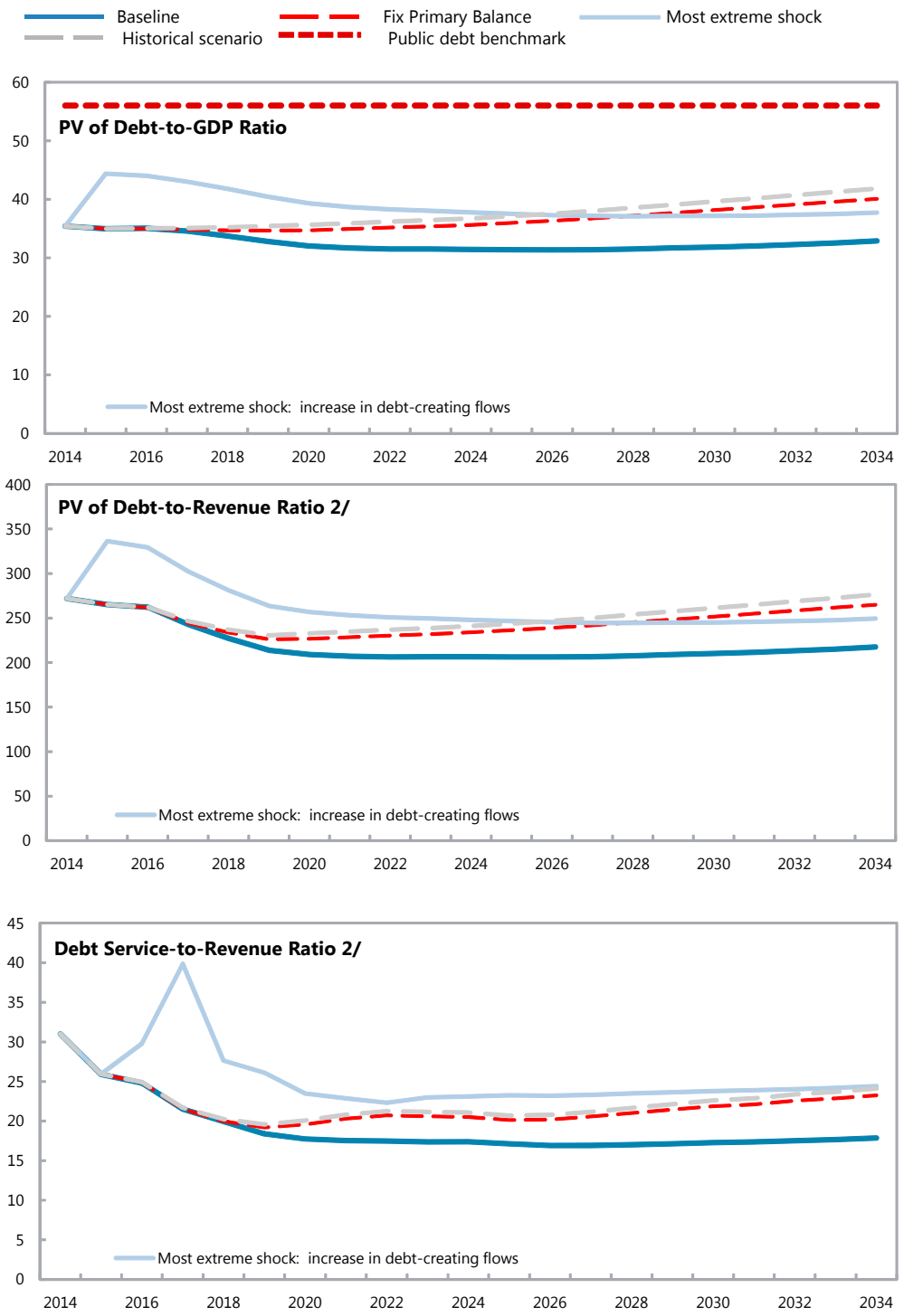


Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a one-time depreciation shock; in c. to a terms shock; in d. to a one-time depreciation shock; in e. to a growth shock and in figure f. to a one-time depreciation shock.

Figure 2. Bangladesh: Indicators of Public Debt under Different Assumptions, 2014-2034 1/

(In percent)



Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

Table 2. Bangladesh: External Debt Sustainability Framework, Baseline Scenario, 2014–2034 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**22.7**	**22.0**	**19.6**			**18.2**	**17.7**	**17.8**	**17.6**	**17.3**	**17.0**		**15.8**	**13.0**	
of which: public and publicly guaranteed (PPG)	21.4	20.6	18.3			17.1	16.6	16.8	16.7	16.4	16.1		14.2	10.6	
Change in external debt	0.3	-0.7	-2.4			-1.4	-0.5	0.1	-0.2	-0.3	-0.3		-0.2	-0.3	
Identified net debt-creating flows	-1.0	-1.4	-5.6			-3.2	-1.4	-1.3	-1.4	-1.3	-1.3		-1.2	-1.7	
Non-interest current account deficit	**1.8**	**0.1**	**-2.2**	**-1.0**	**1.6**	**-1.7**	**0.2**	**0.3**	**0.4**	**0.4**	**0.5**		**0.8**	**1.0**	1.0
Deficit in balance of goods and services	11.8	10.6	7.8			6.1	7.4	7.2	6.9	6.6	6.3		5.4	3.8	
Exports	22.5	23.0	22.6			21.4	19.8	20.5	20.4	20.5	20.5		22.2	27.7	
Imports	34.3	33.6	30.5			27.5	27.3	27.7	27.4	27.1	26.8		27.6	31.5	
Net current transfers (negative = inflow)	-11.1	-11.6	-11.6	-10.0	1.9	-9.4	-8.8	-8.5	-8.1	-7.8	-7.5		-6.2	-4.4	-5.7
of which: official	-0.1	-0.1	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	1.1	1.0	1.5			1.6	1.5	1.6	1.6	1.6	1.6		1.6	1.6	
Net FDI (negative = inflow)	**-0.7**	**-1.0**	**-1.3**	**-1.0**	**0.3**	**-1.0**	**-1.0**	**-1.0**	**-1.1**	**-1.1**	**-1.1**		**-1.6**	**-2.5**	-1.9
Endogenous debt dynamics 2/	**-2.1**	**-0.5**	**-2.1**			**-0.6**	**-0.6**	**-0.7**	**-0.7**	**-0.7**	**-0.7**		**-0.4**	**-0.2**	
Contribution from nominal interest rate	0.2	0.3	0.3			0.3	0.4	0.4	0.4	0.4	0.4		0.5	0.5	
Contribution from real GDP growth	-1.3	-1.4	-1.2			-0.9	-1.0	-1.0	-1.1	-1.1	-1.1		-1.0	-0.8	
Contribution from price and exchange rate changes	-1.0	0.6	-1.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.3**	**0.8**	**3.2**			**1.9**	**0.9**	**1.5**	**1.2**	**1.0**	**1.0**		**1.0**	**1.4**	
of which: exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	13.3			12.6	12.5	12.8	12.8	12.7	12.6		12.3	10.7	
In percent of exports	...	...	58.9			58.7	62.8	62.4	62.8	62.1	61.5		55.2	38.5	
PV of PPG external debt	**...**	**...**	**12.1**			**11.4**	**11.4**	**11.8**	**11.9**	**11.8**	**11.7**		**10.7**	**8.3**	
In percent of exports	**...**	**...**	**53.3**			**53.4**	**57.6**	**57.6**	**58.2**	**57.7**	**57.1**		**48.0**	**30.0**	
In percent of government revenues	**...**	**...**	**97.2**			**91.4**	**89.8**	**91.3**	**86.1**	**81.6**	**78.1**		**71.0**	**55.4**	
Debt service-to-exports ratio (in percent)	**4.1**	**5.7**	**5.9**			**10.7**	**8.8**	**8.3**	**7.2**	**6.9**	**6.4**		**6.3**	**5.0**	
PPG debt service-to-exports ratio (in percent)	**3.8**	**4.8**	**4.4**			**9.3**	**7.4**	**7.1**	**6.0**	**5.7**	**5.3**		**4.7**	**2.9**	
PPG debt service-to-revenue ratio (in percent)	**7.4**	**8.8**	**8.1**			**15.9**	**11.6**	**11.3**	**8.9**	**8.1**	**7.3**		**7.0**	**5.4**	
Total gross financing need (Billions of U.S. dollars)	2.4	1.1	-1.2			0.9	2.7	3.2	2.7	2.7	2.5		2.9	-1.6	
Non-interest current account deficit that stabilizes debt ratio	1.5	0.7	0.2			-0.3	0.8	0.2	0.6	0.8	0.8		1.1	1.3	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.7	6.2	6.0	6.2	0.3	5.5	6.3	6.5	7.0	7.0	7.0	6.6	6.5	6.5	6.5
GDP deflator in US dollar terms (change in percent)	4.5	-2.4	5.5	3.3	4.1	11.7	7.5	2.9	3.3	3.0	3.3	5.3	3.0	3.0	3.0
Effective interest rate (percent) 5/	0.9	1.5	1.4	1.1	0.2	2.1	2.3	2.5	2.6	2.7	2.8	2.5	3.5	4.5	3.8
Growth of exports of G&S (US dollar terms, in percent)	34.5	6.0	10.2	15.1	8.3	11.4	5.9	13.1	10.4	10.6	10.6	10.3	11.6	12.3	12.0
Growth of imports of G&S (US dollar terms, in percent)	50.0	1.6	1.4	15.2	14.9	6.5	13.2	11.3	9.2	9.2	9.2	9.8	10.5	11.2	11.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	23.4	23.7	21.8	22.8	24.3	23.3	23.2	19.0	12.1	16.9
Government revenues (excluding grants, in percent of GDP)	11.7	12.4	12.4			12.5	12.7	12.9	13.8	14.5	15.0		15.0	15.0	15.0
Aid flows (in Billions of US dollars) 7/	1.3	2.3	3.1			1.6	1.7	1.7	1.8	1.9	1.9		2.2	2.7	
of which: Grants	0.3	0.6	0.6			0.8	0.8	0.9	0.9	0.9	0.9		1.0	1.2	
of which: Concessional loans	1.1	1.7	2.4			0.8	0.9	0.9	1.0	1.1	1.1		1.2	1.5	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.0	1.0	0.9	0.8	0.8	0.7		0.5	0.3	0.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			38.1	36.9	35.8	36.0	37.0	35.6		29.8	20.2	26.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	111.9	116.0	129.9			153.0	174.9	191.8	212.0	233.7	258.3		412.5	1044.3	
Nominal dollar GDP growth	11.5	3.7	11.9			17.8	14.3	9.6	10.5	10.2	10.5	12.2	9.7	9.7	9.8
PV of PPG external debt (in Billions of US dollars)			16.1			17.5	20.0	22.3	24.9	27.3	29.9		43.5	86.1	
(PVt-PVt-1)/GDPt-1 (in percent)						1.1	1.6	1.3	1.3	1.1	1.1	1.3	0.8	0.6	0.7
Gross workers' remittances (Billions of US dollars)	11.7	12.7	14.3			13.8	14.6	15.5	16.4	17.4	18.4		24.7	44.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	10.9			10.5	10.6	10.9	11.1	11.0	10.9		10.1	8.0	
PV of PPG external debt (in percent of exports + remittances)	...	...	35.8			37.6	40.6	41.4	42.3	42.3	42.4		37.8	26.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.0			6.6	5.2	5.1	4.4	4.2	4.0		3.7	2.5	

Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ Central government gross debt, including debt owed to the IMF, plus external borrowing by public enterprises that is supported by central government guarantees, including short-term oil-related suppliers' credits. The years in the table refer to fiscal years. For example, 2014 refers to July 2013-June 2014.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3. Bangladesh: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to-GDP+remittances ratio								
Baseline	10	11	11	11	11	11	**10**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	10	10	9	8	7	6	**2**	-1
A2. New public sector loans on less favorable terms in 2014-2034 2	10	11	12	12	13	13	**14**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	10	11	11	11	11	11	**10**	8
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	10	10	11	12	11	11	**10**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	10	11	12	12	12	12	**11**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	10	11	12	12	12	12	**11**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	10	11	11	11	11	11	**10**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	10	14	15	15	15	15	**13**	11
PV of debt-to-exports+remittances ratio								
Baseline	38	41	41	42	42	42	**38**	26
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	38	38	34	32	28	24	**6**	-4
A2. New public sector loans on less favorable terms in 2014-2034 2	38	43	44	47	49	50	**51**	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	38	41	41	42	42	42	**37**	26
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	38	40	45	46	46	46	**41**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	38	41	41	42	42	42	**37**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	38	45	46	46	46	46	**40**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	38	37	37	38	38	38	**35**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	38	41	41	42	42	42	**37**	26
PV of debt-to-revenue ratio								
Baseline	91	90	91	86	82	78	**71**	55
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	91	85	77	65	55	46	**11**	-8
A2. New public sector loans on less favorable terms in 2014-2034 2	91	95	98	96	94	93	**96**	91
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	91	90	91	86	81	78	**71**	55
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	91	89	96	90	85	81	**73**	56
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	91	97	101	96	91	87	**79**	62
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	91	96	101	94	89	85	**75**	56
B5. Combination of B1-B4 using one-half standard deviation shocks	91	92	93	88	84	80	**74**	59
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	91	125	125	118	112	107	**97**	76

Table 3. Bangladesh: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (continued)

(In percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
Debt service-to-exports+remittances ratio								
Baseline	7	5	5	4	4	4	**4**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	7	5	5	4	4	4	**2**	0
A2. New public sector loans on less favorable terms in 2014-2034 2	7	5	5	4	4	4	**4**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	7	5	5	4	4	4	**4**	3
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	7	5	5	5	4	4	**4**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	7	5	5	4	4	4	**4**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	7	5	5	5	4	4	**4**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	7	5	5	4	4	4	**3**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	7	5	5	4	4	4	**4**	3
Debt service-to-revenue ratio								
Baseline	16	12	11	9	8	7	**7**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	16	12	11	9	8	7	**4**	0
A2. New public sector loans on less favorable terms in 2014-2034 2	16	12	11	9	8	7	**7**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	16	12	11	9	8	7	**7**	5
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	16	12	11	9	8	7	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	16	13	13	10	9	8	**8**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	16	12	12	9	8	8	**8**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	16	12	12	9	9	8	**7**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	16	16	16	12	11	10	**10**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	11	11	11	11	11	11	**11**	11

Sources:Bangladesh authorities; IMF and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 4. Bangladesh: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	44.0	43.1	42.3			41.1	40.2	40.1	39.4	38.3	37.2		35.0	35.2	
of which: domestic borrowing by nonfinancial public enterprises	1.8	1.4	0.6			0.6	0.6	0.6	0.6	0.6	0.6		0.6	0.6	
of which: foreign-currency denominated	21.4	20.6	18.3			17.1	16.6	16.8	16.7	16.4	16.1		14.2	10.6	
of which: guarantee-supported external borrowing	...	...	0.1			0.6	1.0	1.3	1.6	1.9	2.1		3.0	4.5	
Change in public sector debt	1.4	-0.9	-0.9			-1.2	-0.9	-0.1	-0.7	-1.0	-1.2		-0.2	0.3	
Identified debt-creating flows	-0.1	-0.5	-1.8			-1.1	-1.1	0.1	-0.7	-1.0	-1.3		-0.2	0.2	
Primary deficit	2.2	1.1	1.6	1.4	0.5	1.4	1.4	1.5	1.1	0.7	0.4	1.1	0.9	1.0	0.8
Revenue and grants	11.9	12.9	12.9			13.0	13.2	13.4	14.2	14.9	15.3		15.2	15.1	
of which: grants	0.3	0.5	0.5			0.5	0.5	0.4	0.4	0.4	0.3		0.2	0.1	
Primary (noninterest) expenditure	14.1	14.1	14.5			14.5	14.6	14.9	15.3	15.5	15.7		16.2	16.1	
Automatic debt dynamics	-2.3	-1.6	-3.6			-3.1	-2.9	-1.8	-2.2	-2.1	-2.1		-1.5	-1.2	
Contribution from interest rate/growth differential	-2.5	-2.3	-1.7			-1.9	-2.1	-1.9	-2.0	-1.9	-1.8		-1.3	-1.0	
of which: contribution from average real interest rate	0.2	0.2	0.7			0.3	0.4	0.6	0.6	0.7	0.7		0.8	1.1	
of which: contribution from real GDP growth	-2.7	-2.6	-2.5			-2.2	-2.4	-2.5	-2.6	-2.6	-2.5		-2.2	-2.1	
Contribution from real exchange rate depreciation	0.2	0.7	-1.8			-1.2	-0.8	0.1	-0.2	-0.2	-0.3		...	...	
Other identified debt-creating flows	0.0	0.0	0.1			0.5	0.4	0.4	0.4	0.4	0.4		0.4	0.4	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.1			0.5	0.4	0.4	0.4	0.4	0.4		0.4	0.4	
Residual, including asset changes	1.6	-0.3	1.0			-0.1	0.2	-0.1	0.0	0.0	0.1		0.0	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	36.0			35.4	35.0	35.1	34.6	33.8	32.8		31.5	32.9	
of which: foreign-currency denominated	...	...	12.1			11.4	11.4	11.8	11.9	11.8	11.7		10.7	8.3	
of which: external	...	...	12.1			11.4	11.4	11.8	11.9	11.8	11.7		10.7	8.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.3	6.4	7.3			6.9	6.0	6.0	5.2	4.6	4.1		4.2	4.1	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	279.2			271.9	265.1	262.3	242.9	227.2	213.9		206.4	217.5	
PV of public sector debt-to-revenue ratio (in percent)	...	...	290.4			283.3	274.8	271.3	250.0	232.9	218.6		209.7	219.3	
of which: external 3/	...	...	97.2			91.4	89.8	91.3	86.1	81.6	78.1		71.0	55.4	
Debt service-to-revenue and grants ratio (in percent) 4/	22.1	24.3	24.6			31.0	25.9	24.8	21.5	20.0	18.4		17.4	17.9	
Debt service-to-revenue ratio (in percent) 4/	22.6	25.3	25.6			32.3	26.9	25.7	22.1	20.5	18.8		17.7	18.0	
Primary deficit that stabilizes the debt-to-GDP ratio	0.7	2.0	2.5			2.7	2.3	1.6	1.8	1.7	1.6		1.1	0.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.7	6.2	6.0	6.2	0.3	5.5	6.3	6.5	7.0	7.0	7.0	6.6	6.5	6.5	6.5
Average nominal interest rate on forex debt (in percent)	0.9	1.2	1.0	1.0	0.1	1.7	2.0	2.1	2.3	2.4	2.5	2.2	2.9	3.5	3.1
Average real interest rate on domestic debt (in percent)	1.8	1.8	4.0	3.3	1.2	1.0	1.6	2.4	2.6	2.8	2.7	2.2	3.4	4.1	3.7
Real exchange rate depreciation (in percent, + indicates depreciation)	1.1	3.5	-9.4	-1.2	5.5	-6.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.5	8.5	6.6	6.6	1.5	8.7	7.5	6.4	6.2	5.9	6.1	6.8	5.2	4.5	4.9
Growth of real primary spending (deflated by GDP deflator, in percent)	21.0	6.1	9.7	3.7	6.9	5.0	7.4	8.4	10.3	8.3	8.5	8.0	5.8	7.2	6.7
Grant element of new external borrowing (in percent)	...	...	...	...	...	23.4	23.7	21.8	22.8	24.3	23.3	23.2	19.0	12.1	...

Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ Central government gross debt including debt owed to the IMF, plus domestic bank borrowing by the nonfinancial public sector and external borrowing by public enterprises that is supported by central government guarantees, including short-term oil-related suppliers' credits. The years in the table refer to fiscal years. For example, 2014 refers to July 2013-June 2014.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 5. Bangladesh: Sensitivity Analysis for Key Indicators of Public Debt, 2014–2034

(In percent)

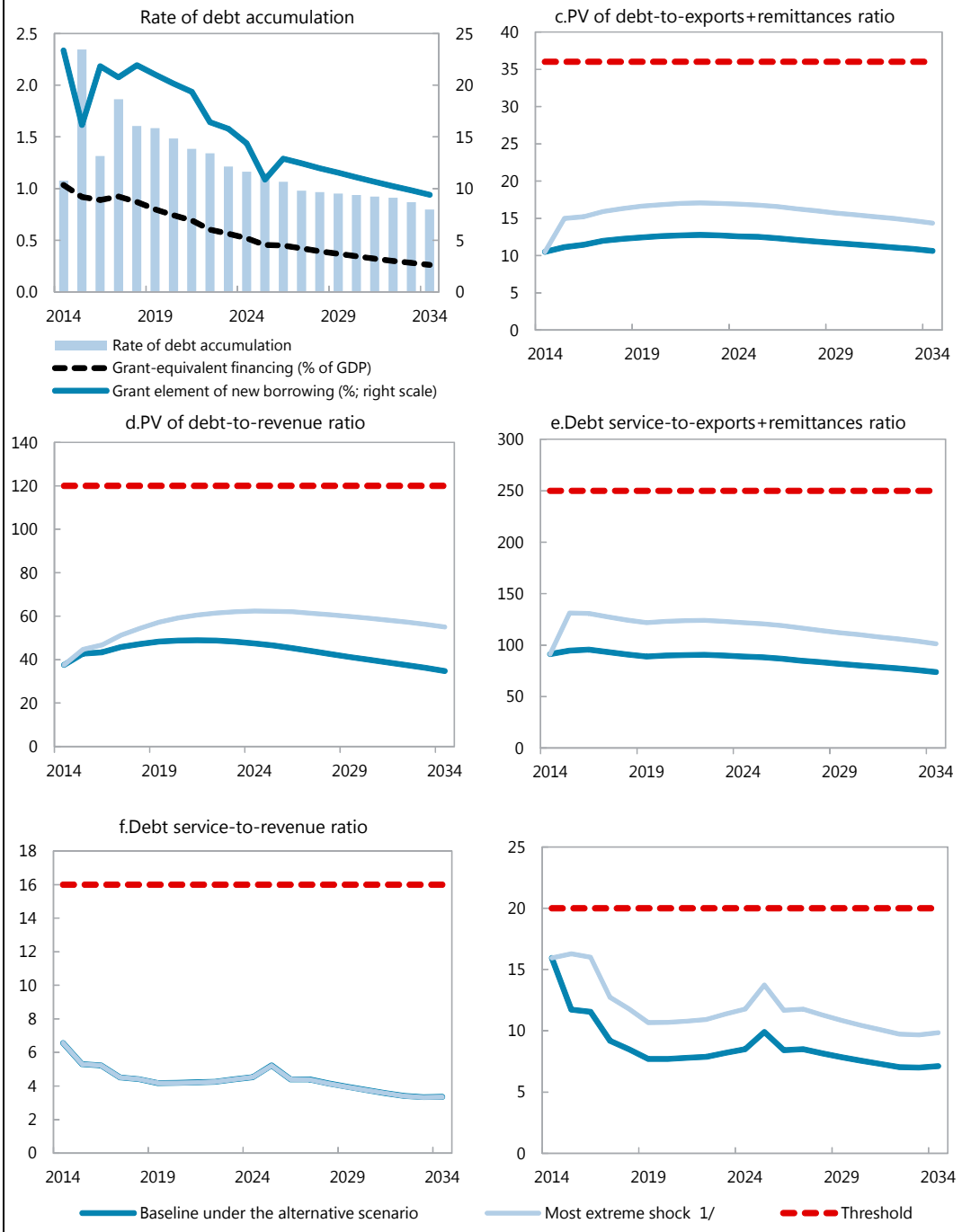
	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of Debt-to-GDP Ratio								
Baseline	35	35	35	35	34	33	**31**	33
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	35	35	35	35	35	35	**37**	42
A2. Primary balance is unchanged from 2014	35	35	35	35	35	35	**36**	40
A3. Permanently lower GDP growth 1/	35	35	35	35	34	33	**32**	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	35	35	36	35	34	34	**33**	35
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	35	35	36	35	34	33	**32**	33
B3. Combination of B1-B2 using one half standard deviation shocks	35	35	36	35	34	34	**32**	34
B4. One-time 30 percent real depreciation in 2015	35	39	39	38	37	36	**35**	36
B5. 10 percent of GDP increase in other debt-creating flows in 2015	35	44	44	43	42	40	**38**	38
PV of Debt-to-Revenue Ratio 2/								
Baseline	272	265	262	243	227	214	**206**	218
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	272	265	262	247	237	231	**241**	276
A2. Primary balance is unchanged from 2014	272	265	262	245	234	226	**234**	265
A3. Permanently lower GDP growth 1/	272	265	263	243	228	215	**210**	229
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	272	266	266	247	231	219	**214**	230
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	272	269	269	248	232	218	**210**	220
B3. Combination of B1-B2 using one half standard deviation shocks	272	267	266	247	232	219	**213**	227
B4. One-time 30 percent real depreciation in 2015	272	298	293	269	251	236	**227**	240
B5. 10 percent of GDP increase in other debt-creating flows in 2015	272	336	329	302	281	264	**248**	250
Debt Service-to-Revenue Ratio 2/								
Baseline	31	26	25	22	20	18	**17**	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	31	26	25	22	20	20	**21**	24
A2. Primary balance is unchanged from 2014	31	26	25	22	20	19	**21**	23
A3. Permanently lower GDP growth 1/	31	26	25	22	20	18	**18**	19
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016	31	26	25	22	20	19	**18**	19
B2. Primary balance is at historical average minus one standard deviations in 2015-2016	31	26	25	23	21	19	**18**	18
B3. Combination of B1-B2 using one half standard deviation shocks	31	26	25	22	21	19	**18**	19
B4. One-time 30 percent real depreciation in 2015	31	28	30	26	25	23	**23**	24
B5. 10 percent of GDP increase in other debt-creating flows in 2015	31	26	30	40	28	26	**23**	24

Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 3. Bangladesh: Alternative Scenario — Indicators of Public and Publicly Guaranteed External Debt under Different Assumptions, 2014-2034 1/

(In percent)

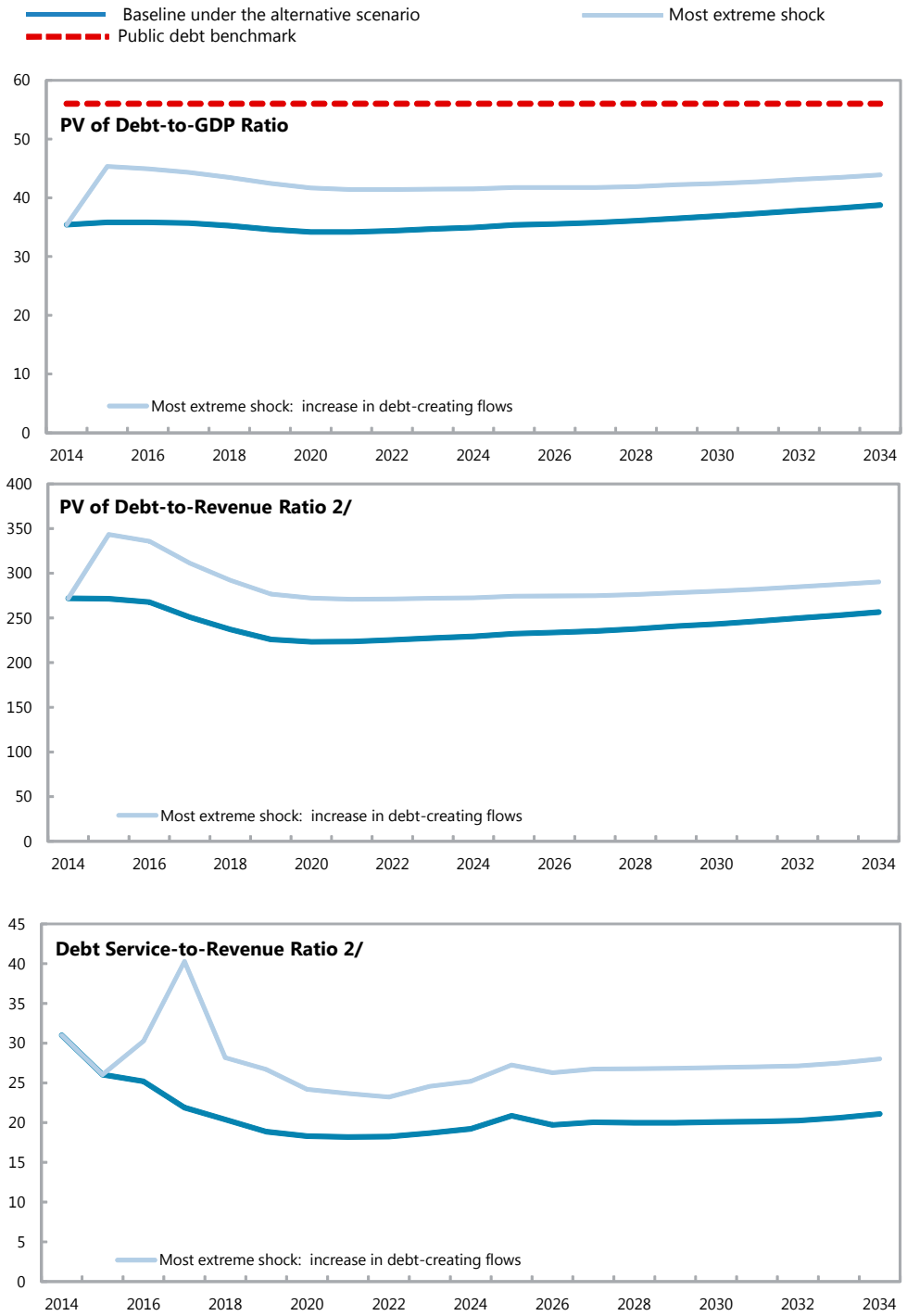


Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a one-time depreciation shock; in c. to a terms shock; in d. to a one-time depreciation shock; in e. to a growth shock and in figure f. to a one-time depreciation shock.

Figure 4. Bangladesh: Alternative Scenario — Indicators of Public Debt under Different Assumptions, 2014-2034 1/

(In percent)



Sources: Bangladesh authorities; and IMF staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.